

Pranay Mathur

+1(404) 9335405 | matnay17@gmail.com | LinkedIn: [pranaymathur17](#) | Website: <https://matnay.github.io/>

EDUCATION

M.S. Robotics (Computer Vision/AI)	Georgia Institute of Technology	Aug 2022 – May 2024
B.E. Electronics and Instrumentation	Birla Institute of Technology and Science, Pilani	Aug 2017 – July 2021

EXPERIENCE

Cobot — Machine Learning Engineer, Perception Seattle, WA	Jan 2025 – Present
- Set up a scalable training, inference, evaluation and monitoring infrastructure to train vision models using sharding and multi-GPU support - Improved model inference time using quantisation-aware training, TensorRT acceleration and CUDA-accelerated processing pipelines - Created a data flywheel to automate data sideling, curation and annotation using knowledge distillation with vision foundation models	
MathWorks — EDG Software Engineer Natick, MA	June 2024 – Jan 2025
- Implemented vision foundation models and wrote custom attention layer implementations to achieve a ~XX% speed up in inference times - Worked on a recommendation system using Word2Vec to match development teams with candidates, achieving a 0.8 F1@10	
MathWorks — Software Intern Natick, MA	May 2023 – Aug 2023
- Developed the test harnesses search feature in the Simulink Test Toolbox using graph search algorithms and deployed it to production - Optimised the C++ and MATLAB back-end of the Simulink Test Toolbox, achieving a 70% speed-up over the original execution time	
Google Summer of Code@CROSS — Software Intern, Open Source Remote	June 2022 – Aug 2022
- Utilised 3D multi-view geometry and object detection (~0.95 mIoU) for mapping landmarks and path-finding for a scale autonomous race car - Implemented full INT8 quantisation and fused layers of EfficientDet to improve inference speed by 10x on an embedded PC with a TPU	
Addverb Technologies — Perception Engineer, Mobile Robotics India	Aug 2021 – July 2022
- Prototyped appearance-based navigation using spatio-temporal LSTM for semantic-scene understanding and efficient image retrieval - Used bag-of-words of learnt key-point descriptors for Visual-Place Recognition (88% recall@1) to augment recovery of SLAM failures - Shipped monitoring and control infrastructure for a fleet of autonomous mobile robots on the cloud for low-latency (<0.1 s) visualisation	
Autonomous Robots Lab, UNR— Undergraduate Researcher Reno, NV	July 2020 - Jan 2021
- Developed a generalizable Resource-Aware algorithm for deployment of Visual Inertial Odometry (VIO) algorithms on computationally constrained aerial vehicles, achieving a reduction in average CPU usage of upto 50% and <0.3 translational MAE and - Released two official ROS and ROS2 perception packages - a ROS wrapper for Open3D and example use-cases with pointclouds	
KPIT Technologies — Research Intern India	May 2020 – July 2020
- Developed CNN-based multi-modal sensor fusion architecture for object detection using a 3D LiDAR, monocular camera and RADAR - Achieved low-latency inference with +4% 3D mIoU over SOTA on the nuScenes dataset in adverse weather, low-illumination and occlusions	

PUBLICATIONS (Selected)

EgoMimic: Scaling Imitation Learning through Egocentric Video

Simar Kareer, Dhruv Patel*, Ryan Punamiya*, **Pranay Mathur***, Shuo Cheng, Chen Wang, Judy Hoffman, Danfei Xu – IEEE International Conference on Robotics and Automation (ICRA), 2025

Neural Visibility Field for Uncertainty-Driven Active Mapping

Shangjie Xue, Jesse Dill, **Pranay Mathur**, Frank Dellaert, P. Tsiotras, Danfei Xu – IEEE/CVF Computer Vision and Pattern Recognition (CVPR), 2024

Proactive Human-Robot Interaction using Visuo-Lingual Transformers and Object Interaction Graphs (*Best Paper Award*)

Pranay Mathur – Geriatrionics Workshop - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023

Resource-aware Online Parameter Adaptation for Computationally-constrained Visual-Inertial Navigation Systems

Pranay Mathur, Nikhil Khedekar, Kostas Alexis - IEEE-RAS International Conference on Advanced Robotics (ICAR), 2021

A Generalized Kalman Filter Augmented Deep-Learning based Approach for Autonomous Landing in MAVs (*Best Paper Award*)

Pranay Mathur, Yash Jangir, Neena Goveas - IEEE International Symposium of ACA on Intelligent Robotics and Industrial Automation (IRIA), 2021

PROJECTS (Selected)

Embodiment Agnostic Long-Horizon Manipulation with Differentiable Kinematics using Human-Play Data	Aug 2023 – May 2024
Faculty Advisor: Dr. Danfei Xu, Assistant Professor at Georgia Tech and Research Scientist at NVIDIA AI	
- Worked on generalizable manipulation policies by scaling imitation learning through egocentric videos of human-play data - Implemented vision encoder for a trajectory prediction model using the DINOv2 foundation model with LoRA. Achieved embodiment agnostic visual representations by aligning latent-space visual embeddings using an auxiliary KL divergence loss and manipulator masking - Implemented a modified version of action-chunking with transformers (ACT) using Differentiable-Kinematics and integrated it with low-level controllers for a modified version of the ALOHA robot platform	
Long-Horizon planning of Next-best-view of NeRFs and Gaussian Splats	Aug 2023 – Nov 2023
Faculty Advisor: Dr. Danfei Xu, Assistant Professor at Georgia Tech and Research Scientist at NVIDIA AI	
- Established SOTA in visibility-based uncertainty quantification in Neural Radiance Fields (NeRFs) applied to active mapping approaches - Implemented pose optimisation pipeline and active mapping baselines to evaluate and compare our proposed approach with current SOTA	
Human-Motion Prediction: With great power comes great res-pose-ability	Jan 2023 – May 2023
Faculty Advisor: Dr. Zsolt Kira, Assistant Professor at the School of Interactive Computing	
- Implemented transformers and Convolutional Seq-to-Seq models for human-motion prediction on computationally-constrained systems - Achieved comparable performance to several baselines implemented in the fairmotion library at reduced computational costs	

SKILLS

Tools/Frameworks: C, C++, Python, Java, Bash, CUDA, PyTorch, TensorFlow, CMake, OpenCV, OpenMP, ROS 1/2, MATLAB, Linux, GitHub, AWS

Courses: Deep Learning, Data Structures and Algorithms, Object Oriented Programming, Advanced Computer Vision, State Estimation and Localization